Geospatial Analysis of Illicit Economies in Post-2021 Myanmar coup d'état Using Nighttime Lights

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By: Jirawat Poongam
MEP 23207
Advisor: Assistant Professor Jun Goto

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Two-year Master's Program of Public Policy
National Graduate Institute for Policy Studies
Tokyo, Japan

ABSTRACT

The 2021 Myanmar coup d'état triggered widespread political and economic disruption. While the country as a whole experienced a decline in economic activity, certain regions saw a surge in illicit activities. Given the limited availability of reliable data in conflict-affected settings, this study employs a Difference-in-Differences (DiD) approach using Nighttime Lights (NTL) data to assess post-coup changes in economic activity, particularly in areas suspected to be linked with transnational organized crimes. National-level analysis indicates an overall decline in NTL following the coup. Moreover, specific locations associated with casinos and online scam centers—such as Shwe Kokko and Me Htaw Tha Lay in Kayin region, along with several areas in Shan region—exhibited statistically significant increases in NTL after the coup compared to other control areas within their respective regions.

Keywords: Myanmar, Coup d'état, Transnational organized crime, Scam centers, Cybercrime, Casinos, Southeast Asia, Night-time light, Difference-in-differences, ACLED, Remote Sensing

CONTENTS

INTRODUCTION

The 2021 Myanmar coup d'état marked a major turning point in both the country's political landscape and governing regime, triggering the nation into civil war and widespread instability. This led to immediate consequences, including both a deepening humanitarian crisis and a sharp economic downturn. Amid the ongoing conflict, illicit activities have surged, including drug and arms trafficking, human smuggling, online scams, and environmentally harmful practices such as unregulated mining and logging. These informal and illegal economies have historically played a role in fueling Myanmar's internal conflicts and political struggles.

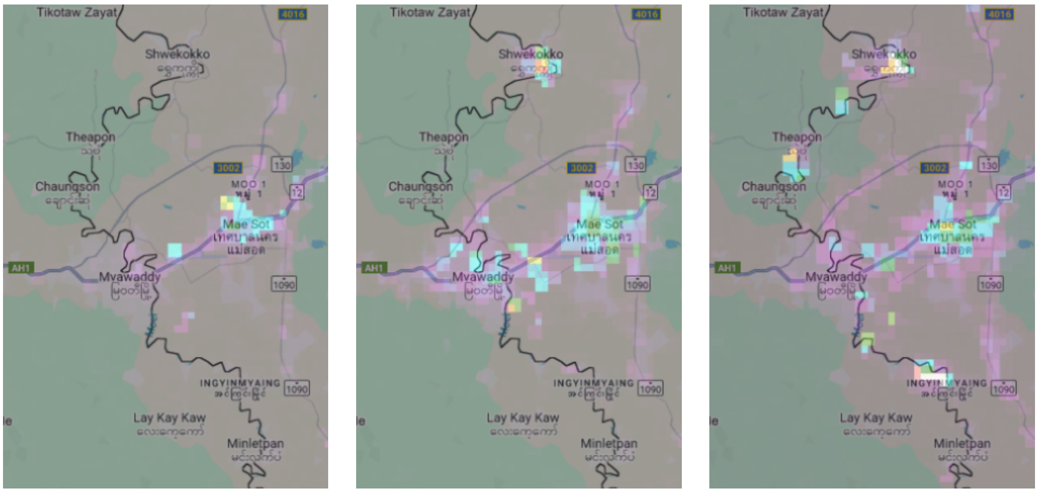
Toxic mining operations, particularly for gold and heavy rare earth elements (HREEs), have surged in response to increasing global demand—especially for components essential to electric vehicles and wind energy. Armed groups across the conflict spectrum—including the military junta, ethnic armed organizations (EAOs), and resistance forces—profit from these extractive industries by controlling areas.[1] The rapid expansion of mining has led to severe environmental degradation, contaminating rivers and posing significant health risks to both people and wildlife due to the use of hazardous chemicals and heavy metals.[2]

The rapid expansion of cyber scam operations, particularly in remote border zones under the control of armed non-state actors. As illustrated by a suspected scam center identified through night-time light emissions captured by the Visible Infrared Imaging Radiometer Suite (VIIRS) in Figure 1, these areas have become hotspots for criminal activity. The 2023 Global Organized Crime Index ranks Myanmar as having the highest level of organized criminality worldwide[3]. Cyberscam activity has become so significant that it affects neighboring countries and globally, leading the U.S. Department of the Treasury's Office of Foreign Assets Control (OFAC) sanctioned the Karen National Army (KNA) and its leader. Estimates indicate Americans are suffering increasing financial losses because of these sophisticated cyber scams from Burma and other Southeast Asian countries, amounting to over $2 billion in 2022 and $3.5 billion in 2023[4].

[1]https://www.reuters.com/world/asia-pacific/myanmar-rebels-disrupt-china-rare-earth-trade-sparking-regional-scramble-2025-03-28/
[2] https://www.dw.com/en/the-dirty-secrets-behind-myanmars-rare-earths-boom/a-72530460
[3] https://globalinitiative.net/analysis/ocindex-2023/
[4]https://mm.usembassy.gov/treasury-sanctions-burma-warlord-and-militia-tied-to-cyber-scam-operations/

Figure 1: Average night-time light in 2015, 2020, and 2024
from VIIRS calculated by NASA (VNP46A3)

2015 | 2020 | 2024



Source: NASA LAADS DAAC

Following the 2021 military coup, Myanmar's overall economy contracted; however, some sectors and areas experienced unexpected growth amid the breakdown of central authority. In many border areas, non-state armed groups and transnational organized crimes expanded their influence. According to the UNODC (2024), areas such as Shwe Kokko in Kayin State and Mongla in Shan State emerged as major centers for casinos, cyber scams, human trafficking, and organized crime, as the locations shown in the figure 2. Investigations have also revealed that military-linked entities expanded control over the mining sector, particularly rare earth minerals[5]. These trends highlight how certain actors and regions have managed to profit from the broader political instability, even as much of the population faces deepening economic hardship.

[5] https://globalwitness.org/en/campaigns/transition-minerals/myanmars-poisoned-mountains/

Figure 2: Locations of known or reported casinos, scam centers and related in Myanmar, 2023

Note: The present map depicts locations of known or reported cyberfraud compounds and related special zones in Myanmar as reported by regional law enforcement authorities and may be subject to change given the evolving situation an ongoing law enforcement operation.
Source: UNODC. (2024). Casinos, Money Laundering, Underground Banking, and Transnational Organized Crime in East and Southeast Asia: A Hidden and Accelerating Threat

Overall, the expansion of unregulated mining and transnational organized crime in Myanmar after the coup has accelerated at unexpected pace, leading to extensive environmental degradation, substantial financial losses, and widespread human rights violations. Understanding the scope and transnational dimensions of these illicit activities is essential—not only for Myanmar's long-term recovery but also for ensuring stability and security across the wider Southeast Asian region.

LITERATURE REVIEW

This section is organized into two parts to review the literature relevant to this study. The first part explores the economic impacts of war and civil conflict, with a specific focus on the Myanmar context. The second part examines the use of night-time lights (NTL) data as a tool for measuring economic activity and monitoring conflict.

The Economic Consequences of War and Civil Conflict

The overall economic impact of war and civil conflict has been widely studied, with the majority of the literature highlighting its destructive effects. Armed conflicts often lead to infrastructure destruction, population displacement, capital flight, and long-term disruptions to trade and investment (Blattman & Miguel, 2010). Civil war, in particular, is associated with a sharp decline in GDP, human development indicators, and state capacity. Collier (1999) found that during the civil war reduces annual GDP growth by about 2.2% on average.

However, recent research has also pointed out that the effects of conflict are not uniformly negative. Some actors and regions may economically benefit from instability. Conflicts often create opportunities for illicit economic activities, rent-seeking behavior, and the capture of natural resource wealth by military or political elites. For example, Le Billon (2001) and Ross (2004) describe how war economies can emerge around the extraction of valuable resources such as gemstones, gold, or timber, particularly in regions outside state control.

In Myanmar, civil conflict has long been a significant obstacle to economic development. Since gaining independence in 1948, prolonged clashes between the central government and ethnic armed organizations have resulted in persistent instability. These regions have suffered from chronic underinvestment and have remained largely detached from the national economy due to ongoing insecurity and weak governance (South, 2008). The country has thus fallen into what is known as a “conflict trap,” where repeated cycles of violence hinder local economic growth, discourage foreign investment, and obstruct nationwide infrastructure development (Collier et al., 2003). In many conflict-affected zones, informal economies—driven by cross-border trade, opium production, and the presence of armed groups—flourished outside the bounds of state control (Woods, 2011).

Using Night-Time Lights (NTL) to Measure Economic Activity

Night-time lights (NTL) satellite data have become an increasingly important tool for estimating economic activity, especially in countries where conventional economic statistics are unreliable or incomplete. Early foundational work by Henderson, et al. (2012) demonstrated the value of NTL in estimating GDP and economic growth across countries. Since then, NTL has been widely adopted to study development, inequality, and urbanization in data-scarce environments.

In conflict settings, Nighttime Lights (NTL) data offer a crucial alternative for monitoring economic activity when traditional field-based data collection is compromised or inaccessible. For example, Witmer and O'Loughlin (2011) using NTL to analyze conflict-related events, including population flows and changes in development, in the Caucasus region of Russia and Georgia. Shortland, et al. (2013) use fine-grained data on night light emission to estimate the economic impacts of violence in Somalia. This is particularly important for countries like Myanmar, where on-the-ground data collection is hindered by conflict, safety risks, and weak governance.

A growing body of research has demonstrated the utility of night-time lights (NTL) data as a proxy for analyzing regional development and economic activity in Myanmar. Kumagai, et al. (2012) used NTL to assess economic disparities across the country, highlighting how luminosity data can effectively approximate subnational GDP in data-scarce environments. Their findings emphasized persistent developmental divides, particularly between central Myanmar and its marginalized borderlands. Bennett and Faxon (2021) use a remote sensing approach to explore the geopolitical and developmental patterns in Myanmar's ethnic minority borderlands. Their study combines Nighttime Lights (NTL) data from 1992 to 2020 with ethnographic research and a comprehensive review of existing literature. The results highlight that, although these regions have experienced faster increases in illumination compared to Myanmar's central lowland areas, they continue to exhibit significantly lower radiance overall. This suggests ongoing spatial inequalities, with borderland areas remaining underdeveloped relative to both national urban centers and adjacent regions across international borders. Moreover, the UNDP Myanmar Development Observatory (2024) found that NTL trends since the 2021 coup reflect regional inequalities and reveal reductions in light emissions across several urban centers, consistent with economic contraction. However, some border regions exhibited stable or even increased luminosity.

Despite advancements in the application of remote sensing to assess economic and conflict-related dynamics, a notable gap remains concerning the emergence of cyber-scam economies in the post-coup period. Areas such as Laukkaing, Pangsang, and Shwe Kokko have increasingly become centers of transnational criminal activity, including human trafficking and illicit online gambling, often operating under weak governance structures (UNODC, 2024). Yet, the spatial and economic footprint of these informal sectors has not been systematically investigated through NTL data, presenting a promising direction for future research at the intersection of remote sensing and political economy in Myanmar.

METHODOLOGY

This study utilizes a difference-in-differences (DiD) approach to examine the rise of illicit activities following the 2021 military coup in Myanmar. The methodology section is divided into two parts. The first part focuses on data collection and preprocessing, detailing the sources of data, descriptive statistics, and the procedures used to prepare the dataset for empirical analysis. The second part outlines the Difference-in-Differences (DiD) analytical framework, including the construction of treatment and control groups, and the assessment of the parallel trends assumption. The study further employs a dynamic DiD (or event study) approach. This method estimates treatment effects across multiple time periods before and after the intervention, allowing for a more detailed examination of pre-treatment trends and the timing of post-treatment responses. Night-time light (NTL) data are employed as a proxy for local economic activity, serving as the primary dependent variable to detect shifts in luminosity patterns potentially associated with the expansion of illicit economies.

Data collection and processing

This study draws on data from three primary sources to investigate the impact of the 2021 Myanmar coup d'état on economic activity and the emergence of illicit operations.

Table 1: Descriptive statistics

Variables	Region	N	Mean	SD	Min	Max
Night-time light (Oct-May: 8M)	Myanmar	3,718	0.325	1.912	0.001	31.713
	Kayin	5,120	0.123	0.362	0.0001	7.803
	Shan	20,934	0.170	1.463	0.0001	133.620
Night-time light (Nov-Apr: 6M)	Myanmar	3,718	0.327	1.939	0.001	32.034
	Kayin	4,913	0.116	0.375	0.0001	7.727
	Shan	20,595	0.173	1.396	0.0001	120.422
Night-time light (Dec-Mar: 4M)	Myanmar	3,718	0.317	1.982	0.0001	32.580
	Kayin	4,280	0.101	0.402	0.0001	8.012
	Shan	19,046	0.165	1.443	0.0001	117.614
ACLED (index)	Myanmar	3,718	19.050	51.417	0	630.5
	Kayin	5,161	0.778	8.905	0	266.0
	Shan	21,138	0.384	5.822	0	511.8
Elevation (Metre)	Myanmar	3,718	432.73	470.26	4.13	2507.34
	Kayin	5,161	233.23	295.31	6.29	1346.51
	Shan	21,138	1024.70	293.89	127.80	1981.50

Note: 1) Night-time light values represent the average from NASA's Black Marble (VNP46A3) during the dry season, shown in digital number units. 2) The ACLED index is calculated similarly to the UNDP's Vulnerability to Conflict Index (VCI), using a weighted average: 75% based on the number of all events (excluding protests) and 25% based on civilian fatalities in each area, as reported by ACLED. 3) Elevation is shown in meters. 4) The "Region" column displays various statistics across Myanmar at the administrative level 3 (ADM3). For Kayin and Shan, a finer administrative level (ADM4) from MIMU is used. 5) For each variable, the table provides the number of observations (N), mean (Mean), standard deviation (SD), minimum, and maximum values.
Source: Level-1 and Atmosphere Archive & Distribution System (LAADS) NASA, the United States Geological Survey (USGS) NASA, Armed Conflict Location & Event Data (ACLED), and Myanmar Information Management Unit (MIMU)

Table 1 summarizes the key variables and their descriptive statistics. For nationwide analysis, administrative boundaries at level 3 (ADM3) from the Global Administrative Areas (GADM) database are used. In contrast, a more granular level of analysis is conducted for Kayin and Shan regions using administrative boundaries at level 4 (ADM4), sourced from the Myanmar Information Management Unit (MIMU).

Night-Time Light (NTL) data from NASA's Black Marble product, specifically the VNP46A3 dataset, was used to assess patterns of human and economic activity across various regions of Myanmar. the VNP46A3 dataset derived from the Visible Infrared Imaging Radiometer Suite (VIIRS) sensor aboard the Suomi NPP satellite. It captures daily observations of nighttime radiance, corrected for atmospheric and lunar illumination effects, cloud cover, and terrain. The VNP46A3 product provides monthly composites of nighttime lights at approximately 500-meter spatial resolution, with radiance values expressed in digital numbers (DN). Unlike earlier NTL sources such as DMSP-OLS, Black Marble includes advanced correction algorithms to minimize noise and enhance detection in low-light, rural, or developing regions, making it highly suitable for fine-grained spatial analysis. To ensure the reliability of the observations and minimize the impact of cloud and weather-related variation, the analysis focuses on dry-season NTL averages. These are calculated across three temporal groupings: an 8-month average (October to May), a 6-month average (November to April), and a 4-month average (December to March).

The Armed Conflict Location & Event Data (ACLED) project is a comprehensive and widely used source of real-time and historical data on political violence and protest events around the world. Each recorded event includes metadata such as date, location, actors involved, number of fatalities, and event type. The data are updated weekly and covered over 100 countries, making it one of the most up-to-date conflict datasets publicly available.

In this research, ACLED data are used to construct a conflict exposure index at the subnational level, serving as a proxy for instability. Specifically, the index draws on the number of reported events (excluding protests) and civilian fatalities, weighted index in each sub-national location using 75% on events and 25% civilian fatalities to reflect their respective impacts on local conditions—mirroring the methodology used in the UNDP's Vulnerability to Conflict Index (VCI)[6].

Digital elevation data from the Shuttle Radar Topography Mission (SRTM) was used to provide geographic information in 2000, such as elevation in each district. NASA JPL contributed this SRTM V3 product, which was obtained via Google Earth Engine. Table 1 data shows that the average elevation in Myanmar is

[6] United Nations Development Programme. (2024). Vulnerability & Conflict Mapping Myanmar's Civilian Vulnerability to Conflict 3 Years after the Military Takeover

432.73 meter above sea level which is higher than Kayin and lower than Shan on average.

Difference-in-differences analysis

The Difference-in-Differences (DiD) approach is a widely used method for estimating causal effects of policy interventions or external shocks—such as the 2021 coup in Myanmar. This technique assesses the impact by comparing the changes in outcomes over time between treatment and control group. The standard DiD regression model is specified as follows:

$$NTL_{it} = \alpha + \beta^1 Post_t + \beta^2 Treated_i + \beta^3 (Post_t \times Treated_i) + \delta Controls_{it} + \epsilon_{it} \quad (1)$$

Where NTL_{it} represents the average night-time light during the dry season in year t for area i, derived from NASA's Black Marble product. $Post_t$ is a dummy variable equal to 1 if year t is after the 2021 Myanmar coup d'état. $Treated_i$ is a dummy variable equal to 1 if area i is identified as a suspected location of illicit activities according to UNODC reports[7]. $Controls_d$ include elevation (measured in meters) and the ACLED index, which is constructed similarly to the UNDP's Vulnerability to Conflict Index (VCI). This index is a weighted average based on 75% of the number of conflict events (excluding protests) and 25% of civilian fatalities, as reported by ACLED.

As this research focuses on the impact of the 2021 Myanmar coup on the growth of illicit activities, treatment locations are identified based on UNODC (2024), which highlights areas with known or suspected casinos, scam centers, and related operations. Prominent hubs for these activities are primarily concentrated in the Kayin and Shan States, including locations such as Shwe Kokko, Myawaddy, Laukkaing (Kokang region), and Mongla. These suspected sites are spatially mapped in Figure 2 and linked to corresponding administrative boundaries, as detailed in Table 2.

The coefficient of interest β^3 captures the causal impact of the 2021 Myanmar coup d'état on suspected illicit activity in designated treatment areas. Specifically, this represents the interaction term between the treatment locations and the post-coup period. A statistically significant and positive β^3 suggests that, nighttime light intensity increased more in treatment areas compared to control areas following the coup, which may indicate a rise in human activity or economic operations—potentially tied to illicit economies.

[7] United Nations Office on Drugs and Crime. (2024). Casinos, Money Laundering, Underground Banking, and Transnational Organized Crime in East and Southeast Asia: A Hidden and Accelerating Threat.

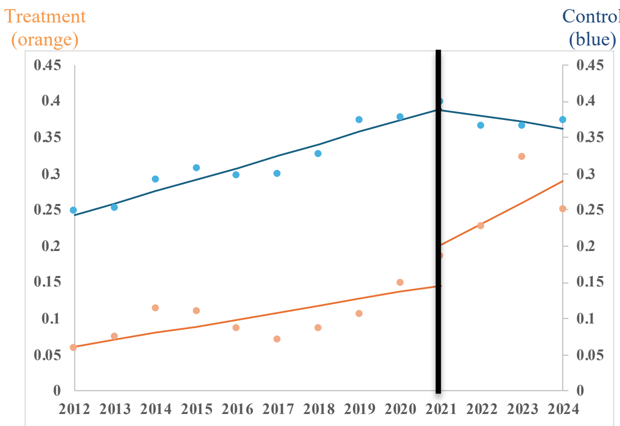
Table 2: Location of known or reported casinos, scam centers and related as treatment areas

	Region	Name	Name of location	Location	Source
ADM3	Kayin	Myawady	Myawadi	MMR.6.3.1_1	GADM
	Shan	Mongla	Keng Tung	MMR.13.1.1_1	GADM
		Mongpauk	Mong Yang	MMR.13.1.2_1	GADM
		Pangsang	Pang-Yang	MMR.13.4.7_1	GADM
		Laukkaing	Kunlong	MMR.13.2.2_1	GADM
ADM4	Kayin	Myawaddy	Me Htaw Tha Lay	MMR003005001	MIMU
			Me Ka Nei	MMR003005009	MIMU
			Myawaddy	MMR003005701	MIMU
			Shewkokko	MMR003005901	MIMU
	Shan	Mongpauk	Mongyang	MMR016003702	MIMU
		Pangsang	Pangsang	MMR015005701	MIMU
		Mongla	Mongla	MMR016005701	MIMU
		Laukkaing	Tar Shwe Htan	MMR015022014	MIMU
			Htin Par Keng	MMR015022002	MIMU
			Shwe Yin See	MMR015022007	MIMU
			Laukkaing	MMR015022701	MIMU

Note: ADM3 refers to the administrative level 3 used for analysis across Myanmar as a whole. For more detailed analysis in specific regions—namely Kayin and Shan—a finer administrative level (ADM4), provided by MIMU, is used.
Source: UNODC (2024), Global administrative areas (GADM) version 2.8 and Myanmar Information Management Unit (MIMU)

The assumption for the internal validity of the Difference-in-Differences model is the parallel trends assumption. This assumption assumes that, in the absence of the coup, the average nighttime light (NTL) in both treatment and control areas would have exhibited similar trends over time. Although this part is not directly tested through statistical means, its credibility can be evaluated by visually inspecting pre-coup trends in NTL data. Figures 3 to 5 illustrate these trends, allowing for an assessment of whether treatment and control areas moved in parallel prior to the intervention.

Figure 3: The average NTL of treatment and control location in Myanmar



Note: The five treatment locations—Myawadi in Kayin State, along with Keng Tung, Mong Yang, Pang Yang, and Kunlong in Shan State—are represented by the orange line, showing the average Nighttime Lights (NTL) during the dry season for each year. In contrast, the 3,713 control locations are represented by the blue line, indicating the average dry-season NTL across the same period. The administrative level 3 used for analysis across Myanmar.
Source: Author's calculation

Figure 4.1: The average NTL between treatment and control location in Kayin

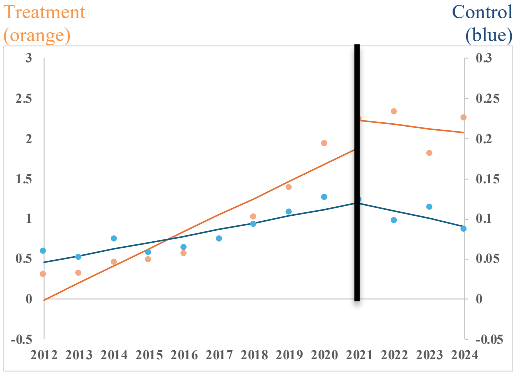
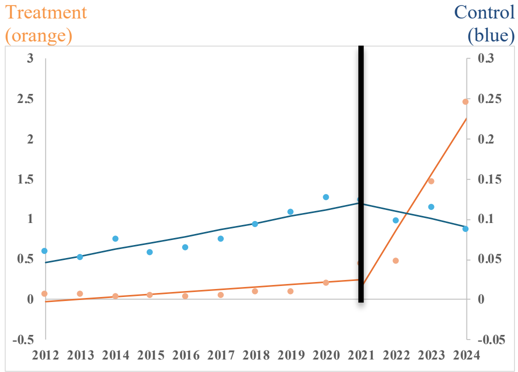
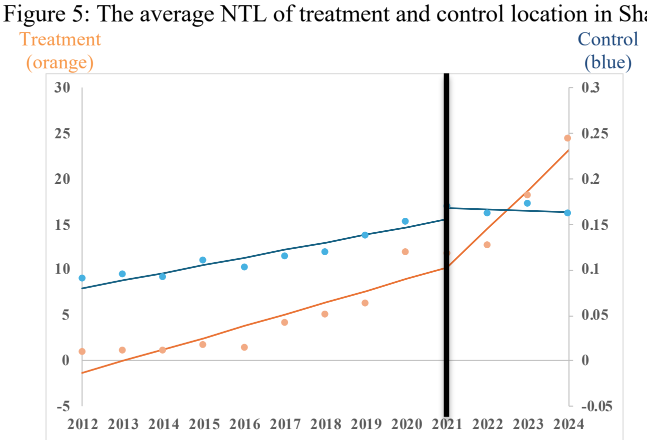


Figure 4.2: The average NTL in Kayin (Treated locations: Shwe Kokko and Me Htaw Tha Lay)



Note: The four treatment locations in Kayin State including Me Htaw Thalay, Me Ka Nei, Myawaddy, and Shewkokko are represented by the orange line in figure 4.1, showing the average Nighttime Lights (NTL) during the dry season for each year. In contrast, the 5,157 control locations in Kayin are represented by the blue line, indicating the average dry-season NTL across the same period. Kayin region used a finer administrative level (ADM4), provided by MIMU.
Source: Author's calculation

Figure 5: The average NTL of treatment and control location in Shan



Note: The seven treatment locations in Shan State including Mongyang, Pangsang, Mongla, Tar Shwe Htan, Htin Par Keng, Shwe Yin See, and Laukkaing are represented by the orange line, showing the average Nighttime Lights (NTL) during the dry season for each year. In contrast, the 21,131 control locations in Kayin are represented by the blue line, indicating the average dry-season NTL across the same period.Shan region used a finer administrative level (ADM4), provided by MIMU.
Source: Author's calculation

Prior to the 2021 coup, both treatment and control areas exhibited a similar upward trend in Night-Time Lights (NTL), consistent with the parallel trends assumption. However, following the coup, a divergence becomes evident: NTL levels in control areas decline, while treatment areas—particularly those suspected of hosting illicit activities—continue to show accelerated growth. However, in the Kayin region (Figure 4.1), the treatment and control groups did not follow parallel trends prior to the coup, suggesting a potential violation of the assumption in this specific region. Nevertheless, when narrowing the focus to newly emerging hubs of illicit activity, such as Shwe Kokko and Me Htaw Tha Lay (also known as KK

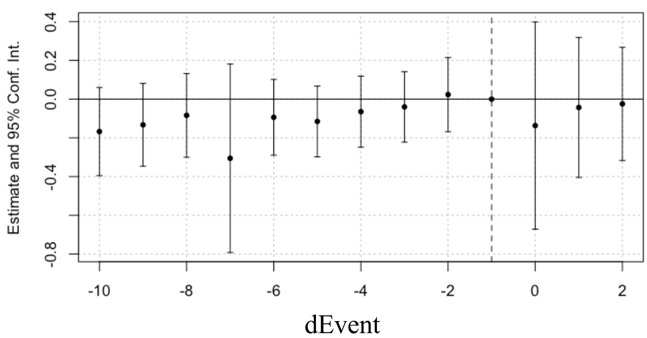
Park), the pre-coup NTL trajectories between treatment and control locations are much more closely aligned. As illustrated in Figure 4.2, these areas probably meet the parallel trends assumption.

This study also adopts a dynamic Difference-in-Differences (DiD) framework, also known as an event study design. Unlike the conventional DiD, which estimates an average treatment effect by comparing outcomes before and after the intervention, the dynamic approach disaggregates the effect across multiple time periods relative to the treatment event. This enables a more detailed examination of pre-treatment trends—crucial for validating the parallel trends assumption—as well as the timing and durability of post-treatment effects. The specification of the dynamic DiD regression model is as follows:

$$NTL_{it} = \alpha + \beta^1 Year_t + \beta^2 Treated_i + \sum_{j\neq-1} \beta_j^3 \left(dEvent_j \times Treated_i\right) + \delta Controls_{it} + \epsilon_{it} \quad (1)$$

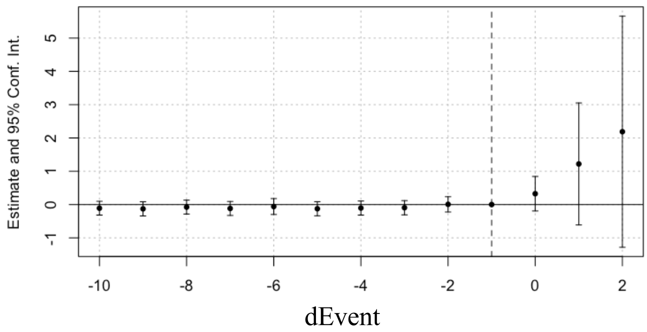
Where $dEvent_j = (Year_t - 2021\text{ Coup})$. The coefficient β_j^3 captures the average treatment effect at each relative time period j, where j < 0 denotes pre-treatment periods and j > 0 denotes post-treatment periods. The period j = -1 is omitted to serve as the reference period, avoiding perfect multicollinearity. To validate the parallel trends assumption, we test whether the coefficients for pre-treatment periods (j < -1) are jointly and individually statistically insignificant. If so, it supports the assumption that, before the event (i.e., the coup), treated and control units would have followed similar trends. A statistically significant β_j for $j \geq 0$ then indicates a treatment effect, and differing values across post-treatment periods $(\beta_j \neq \beta_k)$ suggest a time-varying impact.

Figure 6: Event Study or Dynamic Difference-in-Differences in Myanmar



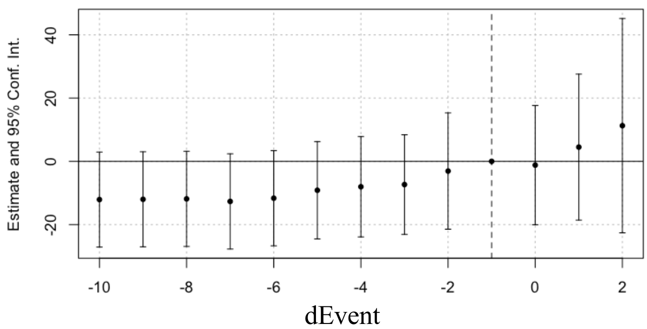
Note: Each point in the graph represents the estimated beta3 coefficient for a given period, accompanied by 95% confidence intervals, illustrating the difference in Nighttime Lights (NTL) between treatment and control locations over time. The study identifies five treatment locations: Myawadi in Kayin State, and Keng Tung, Mong Yang, Pang Yang, and Kunlong in Shan State. In contrast, a total of 3,713 locations serve as control sites. The administrative level 3 used for analysis across Myanmar.
Source: Author's calculation

Figure 7: Event Study or Dynamic Difference-in-Differences in Kayin
(Treated locations: Shwe Kokko and Me Htaw Tha Lay)



Note: Each point in the graph represents the estimated beta3 coefficient for a given period, accompanied by 95% confidence intervals, illustrating the difference in Nighttime Lights (NTL) between treatment and control locations over time. There were two treatment locations in Kayin State including Me Htaw Thalay, and Shewkokko. There were 5,159 control locations in Kayin. Kayin region used a finer administrative level (ADM4), provided by MIMU.
Source: Author's calculation

Figure 8: Event Study or Dynamic Difference-in-Differences in Shan



Note: Each point in the graph represents the estimated beta3 coefficient for a given period, accompanied by 95% confidence intervals, illustrating the difference in Nighttime Lights (NTL) between treatment and control locations over time. There were seven treatment locations in Shan State including Mongyang, Pangsang, Mongla, Tar Shwe Htan, Htin Par Keng, Shwe Yin See, and Laukkaing while the other 21,131 locations served as control.
Source: Author's calculation

To assess the internal validity, the parallel trends assumption is tested by examining the coefficients of pre-treatment periods (j < -1). Figures 6 to 8 show that all β_j are statistically insignificant for j < -1, indicating that treated and control areas exhibited similar NTL trends prior to the coup. This provides support for the parallel trends assumption, suggesting that, before the coup, both groups would have continued comparable trajectories.

RESULTS AND DISCUSSION

This section presents the empirical findings on the impact of the 2021 Myanmar coup d'état on economic activity, human presence, and the expansion of illicit economies, using Night-Time Lights (NTL) data as a proxy. The analysis is structured in two parts to capture both national and regional dynamics. The first part examines patterns across Myanmar at the subnational administrative level 3 (ADM3), offering a broad overview of the country-wide effects. The second part provides a more granular investigation into high-risk regions—specifically Kayin and Shan—using finer-scale administrative level 4 (ADM4). These regions have been identified as hotspots for emerging illicit activities such as unregulated casinos and scam centers.

The impact of the coup on the growth of illicit activities

This part of the analysis investigates nationwide patterns in Myanmar using subnational administrative level 3 (ADM3) boundaries. The study utilizes three types of dry-season Night-Time Lights (NTL) averages to account for seasonal variations: an 8-month average (October–May), a 6-month average (November–April), and a 4-month average (December–March). Following the 2021 coup, Myanmar experienced an average decline of 0.217 digital numbers (DN) in NTL, as observed in regression models 1.7 to 1.9 (Table 3). However, the interaction term for treated areas and the post-coup period (β^3 or Treated Area × Post-Coup) is not statistically significant, indicating that after the coup cannot be conclusively attributed to a rise in illicit activity in these areas. To better capture localized changes potentially linked to illicit economies, the next section focus on finer spatial detail using ADM4-level data in high-risk regions such as Kayin and Shan.

Table 3: The regression results on average night-time light in Myanmar

Y = Average NTL of Myanmar	8M (1.1)	6M (1.2)	4M (1.3)	8M (1.4)	6M (1.5)	4M (1.6)	8M (1.7)	6M (1.8)	4M (1.9)
Treated area	-0.213***	-0.217***	-0.221***	0.054	0.052	0.052	0.051	0.051	0.053
	(0.037)	(0.038)	(0.039)	(0.035)	(0.035)	(0.036)	(0.033)	(0.033)	(0.035)
Post coup d'état	0.051	0.05	0.057	-0.171	-0.176	-0.175	-0.211*	-0.217*	-0.217*
	(0.078)	(0.079)	(0.081)	(0.126)	(0.128)	(0.131)	(0.117)	(0.119)	(0.121)
Treated area*Post coup	0.112	0.101	0.092	0.152	0.142	0.134	0.16	0.149	0.142
	(0.093)	(0.094)	(0.096)	(0.151)	(0.153)	(0.155)	(0.125)	(0.127)	(0.129)
Control variables									
ACLED index				✓	✓	✓	✓	✓	✓
Elevation				✓	✓	✓	✓	✓	✓
Regions							✓	✓	✓
Observations	3718	3718	3718	3718	3718	3718	3718	3718	3718
Adjusted R^2	-0.0005	-0.0010	-0.0005	0.02	0.02	0.02	0.29	0.29	0.29

Note: The analysis covers three types of averaging night-time light during dry season: an 8-month average (October-May), a 6-month average (November-April) and a 4-month average (December-March). Each column presents coefficient estimates with *** p<0.01, ** p<0.05, and * p<0.1, Heteroskedasticity-robust standard errors in parentheses, number of observations, and adjusted R-squared values.
Source: Author's calculation

The impact of the coup on the growth of illicit activities in Kayin and Shan

In the Kayin region, the treatment and control groups did not exhibit parallel trends before the 2021 coup, indicating a potential violation of the DiD assumption. However, when focusing solely on newly emerging illicit hubs—Shwe Kokko and Me Htaw Tha Lay—the pre-coup trends align more closely. Thus, while results are presented for both all treatment areas (Table 4.1) and the restricted subset (Table 4.2), the primary interpretation and discussion will be based on Table 4.2, where the parallel trends assumption is not violated.

Table 4.1: The regression results on average night-time light in Kayin region

Y = Average NTL of Kayin	8M (2.1)	6M (2.2)	4M (2.3)	8M (2.4)	6M (2.5)	4M (2.6)	8M (2.7)	6M (2.8)	4M (2.9)
Treated area	0.886***	0.940***	0.871***	0.875***	0.929***	0.855***	0.876***	0.930***	0.869***
	(0.291)	(0.305)	(0.293)	(0.287)	(0.301)	(0.287)	(0.287)	(0.301)	(0.287)
Post coup d'état	0.015	0.019*	0.017	-0.005	0.001	-0.008	-0.005	0.001	-0.01
	(0.010)	(0.010)	(0.012)	(0.010)	(0.011)	(0.012)	(0.010)	(0.010)	(0.012)
Treated area*Post coup	1.178	1.102	1.155	0.763	0.694	0.586	0.753	0.683	0.577
	(0.753)	(0.750)	(0.753)	(0.590)	(0.594)	(0.569)	(0.590)	(0.593)	(0.569)
Control variables									
ACLED index				✓	✓	✓	✓	✓	✓
Elevation				✓	✓	✓	✓	✓	✓
Provinces							✓	✓	✓
Observations	5120	4913	4280	5120	4913	4280	5120	4913	4280
Adjusted R^2	0.12	0.12	0.11	0.17	0.16	0.17	0.18	0.18	0.19

Note: The analysis covers three types of averaging night-time light during dry season: an 8-month average (October-May), a 6-month average (November-April) and a 4-month average (December-March). Each column presents coefficient estimates with *** p<0.01, ** p<0.05, and * p<0.1, Heteroskedasticity-robust standard errors in parentheses, number of observations, and adjusted R-squared values.
Source: Author's calculation

Table 4.2: The regression results on average night-time light in Kayin region
(Treated locations: Shwe Kokko and Me Htaw Tha Lay)

Y = Average NTL of Kayin	8M (3.1)	6M (3.2)	4M (3.3)	8M (3.4)	6M (3.5)	4M (3.6)	8M (3.7)	6M (3.8)	4M (3.9)
Treated area	0.013	0.019	0.02	-0.006	0.0005	-0.005	-0.189***	-0.189***	-0.162***
	(0.028)	(0.033)	(0.035)	(0.015)	(0.017)	(0.017)	(0.062)	(0.065)	(0.062)
Post coup d'état	0.02	0.024*	0.023	-0.011	-0.007	-0.018	-0.01	-0.005	-0.018
	(0.013)	(0.014)	(0.016)	(0.011)	(0.011)	(0.013)	(0.011)	(0.011)	(0.013)
Treated area*Post coup	1.284**	1.293**	1.319**	1.339**	1.348**	1.393**	1.338**	1.346**	1.392**
	(0.623)	(0.622)	(0.640)	(0.621)	(0.620)	(0.638)	(0.622)	(0.621)	(0.639)
Control variables									
ACLED index				✓	✓	✓	✓	✓	✓
Elevation				✓	✓	✓	✓	✓	✓
Provinces							✓	✓	✓
Observations	5120	4913	4280	5120	4913	4280	5120	4913	4280
Adjusted R^2	0.02	0.02	0.02	0.10	0.09	0.11	0.12	0.12	0.14

Note: The analysis covers three types of averaging night-time light during dry season: an 8-month average (October-May), a 6-month average (November-April) and a 4-month average (December-March). Each column presents coefficient estimates with *** p<0.01, ** p<0.05, and * p<0.1, Heteroskedasticity-robust standard errors in parentheses, number of observations, and adjusted R-squared values.
Source: Author's calculation

Following the 2021 military coup, Kayin region did not exhibit a statistically significant decline in average nighttime light (NTL) levels. However, the interaction term between treated areas and the post-coup period shows a statistically significant increase of approximately 1.3 to 1.4 digital numbers. This suggests a rise in human and economic activity in designated treatment zones, particularly in areas such as Shwe Kokko and Me Htaw Tha Lay (KK Park), which have been identified transnational organized crime locations by UNODC (2024).

In the Shan region, the period following the 2021 military coup is associated with a statistically significant increase in average nighttime light (NTL) levels, diverging from the national trend and from patterns documented in prior conflict-related studies. However, the interaction term for treated areas and the post-coup period (β^3: Treated Area×Post-Coup) is statistically significant, reflecting an average increase of approximately 1.4 digital numbers. This indicates heightened human and economic activity on average in Shan region after the coup, with significant growth in zones identified as hubs of illicit activity.

Table 5: The regression results on average night-time light in Shan region

Y = Average NTL of Shan	8M (4.1)	6M (4.2)	4M (4.3)	8M (4.4)	6M (4.5)	4M (4.6)	8M (4.7)	6M (4.8)	4M (4.9)
Treated area	4.411***	4.253***	4.267***	4.312***	4.153***	4.165***	6.043***	5.817***	5.929***
	(1.279)	(1.205)	(1.153)	(1.288)	(1.214)	(1.161)	(1.502)	(1.408)	(1.354)
Post coup d'état	0.055***	0.053***	0.047***	0.046***	0.044***	0.039***	0.045***	0.043***	0.038***
	(0.011)	(0.011)	(0.012)	(0.011)	(0.011)	(0.012)	(0.011)	(0.012)	(0.013)
Treated area*Post coup	14.370**	13.630**	13.887**	14.383**	13.643**	13.902**	14.384**	13.644**	13.925**
	(7.161)	(6.630)	(6.588)	(7.162)	(6.630)	(6.588)	(6.868)	(6.341)	(6.285)
Control variables									
ACLED index				✓	✓	✓	✓	✓	✓
Elevation				✓	✓	✓	✓	✓	✓
Provinces							✓	✓	✓
Observations	20934	20595	19046	20934	20595	19046	20934	20595	19046
Adjusted R^2	0.20	0.20	0.21	0.20	0.20	0.21	0.24	0.25	0.25

Note: The analysis covers three types of averaging night-time light during dry season: an 8-month average (October-May), a 6-month average (November-April) and a 4-month average (December-March). Each column presents coefficient estimates with *** p<0.01, ** p<0.05, and * p<0.1, Heteroskedasticity-robust standard errors in parentheses, number of observations, and adjusted R-squared values.
Source: Author's calculation

To conclude, this analysis investigates the impact of Myanmar's 2021 military coup on nighttime light (NTL) patterns across multiple administrative levels. At the national scale (ADM3), Myanmar exhibited an average decline in NTL (−0.217 DN), and the interaction term for treated areas with post-coup was not statistically significant, indicating no clear evidence of increased in transnational organized crime (TOC) after the coup at this level of analysis. Therefore, the study shifts focus to finer or ADM4-level data in high-risk regions. In Kayin, when specify only emerging hubs including Shwe Kokko and Me Htaw Tha Lay as treatment locations, there was a statistically significant increase in NTL (1.3–1.4 DN) after the coup compared to control areas, suggesting a surge in human and economic activity potentially tied to TOC. Shan region showed a statistically significant increase in average NTL following the coup, diverging from the national trend. However, treatment zones in Shan experienced significantly greater increases in NTL compared to control areas, indicating heightened human and economic activity on average in Shan region after the coup, with significant growth in zones identified as hubs of illicit activity.

Discussion

This study examines the impact of Myanmar's 2021 military coup on economic activity using Night-Time Lights (NTL) as a proxy. At the national level, results show an average decline in NTL following the coup —consistent with previous literature. However, using ADM3 unit covers a relatively large boundary area, often covering unrelated to the specific treatment zones, which limits the ability to detect effects associated with targeted or illicit economic activities.

The results reveal the clearer patterns when the analysis shifts focus to finer-scale ADM4-level in Kayin and Shan. In Kayin State, newly emerging hubs such as Shwe Kokko and Me Htaw Tha Lay experienced significant increases in NTL post-coup. However, this effect disappears when including urban Myawaddy (MMR003005701), a long-established border trade center. Myawaddy's pre-coup economic development likely violated the parallel trends assumption required for a difference-in-differences approach. Moreover, the area experienced substantial post-coup instability. According to ACLED, Myawaddy recorded 111, 102, and 65 conflict incidents in 2022, 2023, and 2024 respectively, with fatalities totaling 254, 263, and 116 which is high when compared to other areas in Kayin.

Shan State exhibited a statistically significant increase in NTL after the coup, particularly in areas suspected to be involved in illicit activities. Locales such as Pangsang (MMR015005701), Mong Pauk (MMR016003702), and Mongla (MMR016005701)—all under the influence of the United Wa State Army (UWSA). These areas remained relatively insulated from post-coup conflict, with few incidents reported by ACLED. However, Laukkaing township (MMR015022701), which was heavily affected by conflict during Operation 1027 in late 2023, aligning with patterns observed in conflict-affected Myawaddy.

These findings underscore the importance of granular analysis when evaluating the economic impacts of political shocks. While national-level averages indicate an overall economic decline, subnational results highlight significant heterogeneity—shaped by local governance structures, varying degrees of conflict exposure, and the presence of illicit economic activity.

CONCLUSION AND POLICY IMPLICATION

This study highlights the uneven economic impact of Myanmar's 2021 military coup, using night-time lights (NTL) as a proxy for economic activity. While national-level NTL data suggest an overall economic decline, more granular analysis at the ADM4 level reveals significant increases in luminosity in specific border regions after the coup such as Shwe Kokko and Me Htaw Tha Lay in Kayin, and various sites of Shan compared to other control areas. These transnational organized crime locations are consistent with the reported proliferation of cyber scams, illegal trade, and other unregulated activities. The findings underscore the value of satellite-based NTL as a cost-effective, scalable proxy for monitoring economic activity in data-constrained environments. It also offers a powerful tool for detection of hotspots associated with organized crime, particularly when ground data is inaccessible or politically sensitive.

The use of Nighttime Lights (NTL) data presents a valuable opportunity for policymakers to monitor not only the overall economic impact of conflict at the national level but also to uncover spatial disparities and localized changes in activity. This capability is especially valuable in areas where traditional data collection is limited. This spatial lens enables more precise and timely interventions. The observed patterns underscore the urgent need for targeted monitoring and enforcement in newly emerging hubs of illicit activity. Southeast Asian governments and related should focus on enhancing regional collaboration and investing in data-informed monitoring systems that incorporate remote sensing technologies —such as Nighttime Lights (NTL)—with ground-level intelligence to improve the effectiveness of humanitarian assistance, governance, and security interventions.

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